TraceMind AI - Forensic Investigation Report

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1. Executive Summary

2. Correlated Findings & Evidence (9)

#1: Multiple Authentication Failures Preceding Successful Logon (7 fails) [HIGH]

Confidence: ANALYTICAL_FINDING | Category: AUTH_ANOMALY | Time Window: 600s -> 635s

Forensic Reasoning:

Host 'C553' received 7 failed logon attempts within 300s before a successful logon by user 'U999@DOM1'. This temporal sequence is consistent with credential guessing, password spraying, or misconfiguration.

Involved Hosts: C17693, C553 | Involved Users: U999@DOM1

Evidence Sources: auth.txt.gz:L31, auth.txt.gz:L30, auth.txt.gz:L29, auth.txt.gz:L34, auth.txt.gz:L28

Defensive Recommendation: Review the source IP/host, verify if the user legitimately attempted login, and inspect for secondary access tokens.

#2: Multiple Authentication Failures Preceding Successful Logon (6 fails) [HIGH]

Confidence: ANALYTICAL_FINDING | Category: AUTH_ANOMALY | Time Window: 600s -> 720s

Forensic Reasoning:

Host 'C800' received 6 failed logon attempts within 300s before a successful logon by user 'U999@DOM1'. This temporal sequence is consistent with credential guessing, password spraying, or misconfiguration.

Involved Hosts: C17693, C800, C553 | Involved Users: U999@DOM1

Evidence Sources: auth.txt.gz:L31, auth.txt.gz:L30, auth.txt.gz:L29, auth.txt.gz:L28, auth.txt.gz:L27

Defensive Recommendation: Review the source IP/host, verify if the user legitimately attempted login, and inspect for secondary access tokens.

#3: Suspicious Process Execution Followed Remote Authentication [HIGH]

Confidence: ANALYTICAL_FINDING | Category: PROCESS_ANOMALY | Time Window: 635s -> 725s

Forensic Reasoning:

Immediately after user 'U999@DOM1' logged onto 'C553' at t=635, 3 command execution / reconnaissance process(es) were launched (cmd.exe, powershell_recon.exe, vssadmin.exe).

Involved Hosts: C800, C553, C17693 | Involved Users: U999@DOM1

Evidence Sources: auth.txt.gz:L32, proc.txt.gz:L26, proc.txt.gz:L27, proc.txt.gz:L28

Defensive Recommendation: Isolate endpoint, inspect command-line arguments and parent process tree, and capture volatile memory if warranted.

#4: Suspicious Process Execution Followed Remote Authentication [HIGH]

Confidence: ANALYTICAL_FINDING | Category: PROCESS_ANOMALY | Time Window: 720s -> 725s

Forensic Reasoning:

Immediately after user 'U999@DOM1' logged onto 'C800' at t=720, 1 command execution / reconnaissance process(es) were launched (vssadmin.exe).

Involved Hosts: C800, C553 | Involved Users: U999@DOM1

Evidence Sources: auth.txt.gz:L33, proc.txt.gz:L28

Defensive Recommendation: Isolate endpoint, inspect command-line arguments and parent process tree, and capture volatile memory if warranted.

#5: Large Outbound Flow Preceded by DNS Resolution (Potential Data Exfiltration) [CRITICAL]

Confidence: HYPOTHESIS | Category: EXFILTRATION | Time Window: 658s -> 660s

Forensic Reasoning:

Host 'C553' resolved 'C9999_EXFIL' at t=658, followed by a high-volume network flow of 98,500,000 bytes to 'C9999_EXFIL' at t=660.

Involved Hosts: C553, C9999_EXFIL | Involved Users: None

Evidence Sources: dns.txt.gz:L26, flows.txt.gz:L26

Defensive Recommendation: Inspect network capture, examine destination endpoint reputation, and check data transfer authorization.

#6: Rapid Multi-Host Logons for User U103@DOM1 (Potential Lateral Movement) [HIGH]

Confidence: ANALYTICAL_FINDING | Category: LATERAL_MOVEMENT | Time Window: 100s -> 332s

Forensic Reasoning:

User 'U103@DOM1' successfully logged into 4 distinct hosts (C101, C103, C102, C200) within 300s.

Involved Hosts: C101, C103, C102, C200 | Involved Users: U103@DOM1
Evidence Sources: auth.txt.gz:L1, auth.txt.gz:L3, auth.txt.gz:L4, auth.txt.gz:L5, auth.txt.gz:L7
Defensive Recommendation: Check if this account is a service account or domain administrator, and verify if multi-system hop is legitimate.

#7: Rapid Multi-Host Logons for User U102@DOM1 (Potential Lateral Movement) [HIGH]

Confidence: ANALYTICAL_FINDING | Category: LATERAL_MOVEMENT | Time Window: 123s -> 354s

Forensic Reasoning:
User 'U102@DOM1' successfully logged into 4 distinct hosts (C101, C103, C102, C200) within 300s.
Involved Hosts: C101, C103, C102, C200 | Involved Users: U102@DOM1
Evidence Sources: auth.txt.gz:L2, auth.txt.gz:L8, auth.txt.gz:L9, auth.txt.gz:L12, auth.txt.gz:L13
Defensive Recommendation: Check if this account is a service account or domain administrator, and verify if multi-system hop is legitimate.

#8: LANL Ground Truth Red Team Activity Matched (Host: C553) [CRITICAL]

Confidence: CONFIRMED_GROUND_TRUTH | Category: GROUND_TRUTH | Time Window: 600s -> 725s

Forensic Reasoning:
Official LANL red-team log confirms attack simulation at relative time t=635 involving User: 'U999@DOM1', Source: 'C17693', Destination: 'C553'. Found 13 corresponding system event(s) in observed logs within 300s.
Involved Hosts: C800, C553, C17693 | Involved Users: U999@DOM1
Evidence Sources: redteam.txt.gz:L1, auth.txt.gz:L26, auth.txt.gz:L27, auth.txt.gz:L28, auth.txt.gz:L34
Defensive Recommendation: Compare defense detection rules against this confirmed baseline to calibrate sensitivity and false-negative rates.

#9: LANL Ground Truth Red Team Activity Matched (Host: C800) [CRITICAL]

Confidence: CONFIRMED_GROUND_TRUTH | Category: GROUND_TRUTH | Time Window: 600s -> 725s

Forensic Reasoning:
Official LANL red-team log confirms attack simulation at relative time t=720 involving User: 'U999@DOM1', Source: 'C553', Destination: 'C800'. Found 15 corresponding system event(s) in observed logs within 300s.
Involved Hosts: C9999_EXFIL, C800, C553, C17693 | Involved Users: U999@DOM1
Evidence Sources: redteam.txt.gz:L2, auth.txt.gz:L26, auth.txt.gz:L27, auth.txt.gz:L28, auth.txt.gz:L29
Defensive Recommendation: Compare defense detection rules against this confirmed baseline to calibrate sensitivity and false-negative rates.

3. Key Evidentiary Event Timeline (Top 30)

4. Chain of Custody & Cryptographic Provenance

5. Hypotheses, Alternative Explanations & Limitations

Plausible Hypotheses:
- Credential Guessing / Brute-Force: An unauthorized entity systematically tested credentials until achieving valid access.
- Post-Exploitation Execution: The actor obtained interactive command shell execution to execute discovery tools.
- Data Exfiltration: Sensitive data staged on the host was transmitted outbound over network sockets.

Alternative / Benign Explanations:
- Benign Misconfiguration: A scheduled service, script, or user with expired credentials repeatedly retried automated authentication.
- Legitimate Administrative Activity: A system administrator or DevOps automation script executed maintenance tasks.
- Routine Backup / File Synchronization: Legitimate file transfers or database synchronization occurred.

Investigation Boundaries & Limitations:
- Dataset De-identification: LANL dataset uses masked identifiers (e.g., C17693, U620@DOM1) without public IP addresses or full hostnames.
- Process Arguments Missing: Process logs indicate executable names (e.g., cmd.exe) but lack full command-line arguments and parent process GUIDs.
- Temporal Relativity: Timestamps are preserved as relative integer seconds from dataset epoch rather than calendar UTC dates.
- Sampling Boundaries: Any row limits or time sampling configured in ingestion constrain the visibility of precursor or follow-on events.
- Unsupervised ML Boundaries: Isolation Forest scores highlight statistical outliers; they do not prove malicious intent or compromise.